



Loan Collectability & Recovery Analytics Toolbox

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Target Stakeholders: Recovery Operations & Strategy (1st & 2nd Line of Defense)

A Streamlit-based interactive analytics dashboard that can be used for modeling **loan collectability**, **expected recovery value**, and **portfolio-level cash-flow projections**. This app allows analysts to upload real data or generate synthetic demo portfolios, visualize recovery potential, view goals, expected recoveries, and export results for further analysis.



Features

◆ 1. Synthetic or Uploaded Data

- Upload your own **CSV or Excel** file
OR
- Generate a synthetic portfolio with adjustable account count

◆ 2. Recovery Probability & NPV Modeling

The app computes:

- Recovery probability (based on delinquency & strategy)
- Expected Net Present Value (NPV)
- Estimated recovery month
- Delinquency bucket classification

◆ 3. Interactive Visualizations

Includes:

- **Recovery Map (Bubble Chart)** — NPV vs. recency with trendline
- **12-Month Projected Cash Flow**
- **Bucket Concentration Bar Chart**

- Goal Progress Gauge

◆ 4. Account Ledger & Export

- Sortable, formatted account table
- Export full results to CSV
- Downloadable Excel template with sample data

◆ 5. Priority Action Items

Automatically identifies the **Top 5 accounts** with highest expected recovery value.

Installation

```
# Clone the repository
git clone https://github.com/your-repo/loan-collectability-toolbox.git
cd loan-collectability-toolbox
```

```
# Install dependencies
pip install -r requirements.txt
```

```
streamlit
pandas
numpy
plotly
xlsxwriter
```

Screenshots

Data Source

Select Source:

Synthetic Demo

Upload Data

Number of Accounts

300

Recovery Goals

Set Recovery Target (\$)

100000

-

+

Target: \$100,000

Strategy

Discount Rate (%)

8.00

Scenario:

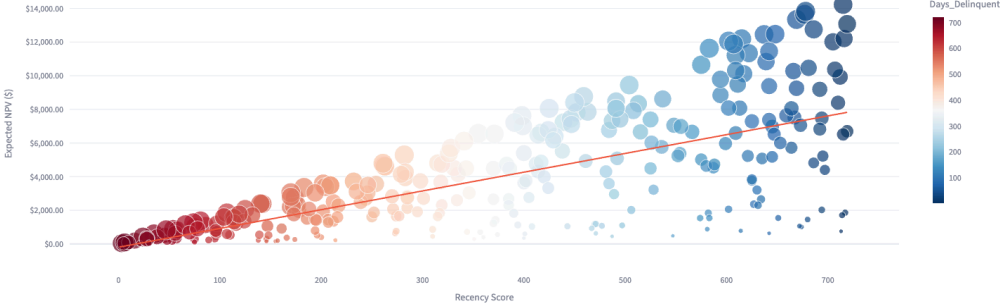
Standard

Aggressive

Conservative

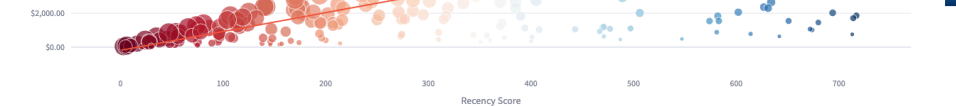
Loan Collectability & Recovery Analytics

Portfolio Recovery Map



Detailed Account Ledger

Manage app



Detailed Account Ledger

Account_ID	Debt Amount	Days_Delinquent	Recovery Prob	Expected NPV	Status
L-97538	\$14,345.55	5	1%	\$14,230.32	Contacted
L-80271	\$14,815.50	42	1%	\$13,823.44	New
L-92798	\$14,649.86	43	1%	\$13,645.73	In Negotiation
L-95305	\$14,516.49	51	1%	\$13,338.33	Contacted
L-57254	\$13,094.69	1	1%	\$13,073.63	In Negotiation
L-71087	\$13,485.09	34	1%	\$12,752.91	Promise to Pay
L-94654	\$14,074.24	72	1%	\$12,468.52	In Negotiation
L-72955	\$14,331.96	83	1%	\$12,451.25	New
L-96202	\$12,285.59	4	1%	\$12,206.64	New
L-10860	\$14,608.65	105	1%	\$12,194.36	In Negotiation

https://github.com/gram99/loan_identifier_poc/blob/main/README.md

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12-Month Projected Cash Flow

Month	NPV Value
May 2026	\$250,000
Jun 2026	\$245,000
Jul 2026	\$105,000
Aug 2026	\$135,000
Sep 2026	\$105,000
Oct 2026	\$105,000
Nov 2026	\$70,000
Dec 2026	\$75,000
Jan 2027	\$55,000
Feb 2027	\$40,000
Mar 2027	\$30,000
Apr 2027	\$10,000

Bucket Concentration

Bucket	Debt Amount
0-30	\$100,000
31-60	\$100,000
61-90	\$100,000
91-180	\$300,000
181-360	\$600,000
361-720	\$1,200,000

Recovery Goal Progress

Month	NPV Value
May 2026	\$100,000
Jun 2026	\$100,000
Jul 2026	\$100,000
Aug 2026	\$100,000
Sep 2026	\$100,000
Oct 2026	\$100,000
Nov 2026	\$70,000
Dec 2026	\$70,000
Jan 2027	\$50,000
Feb 2027	\$40,000
Mar 2027	\$30,000
Apr 2027	\$10,000

Top 10 Expected Recoveries

Bucket	Expected Recovery
0-30	\$100,000
31-60	\$100,000
61-90	\$100,000
91-180	\$300,000
181-360	\$400,000
361-720	\$400,000

Recovery Goal Progress

Value
1.2M
1.1M (Target)

Top 10 Expected Recoveries

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